



SOLUTIONS – LEVEL 4 (JEE ADVANCED STYLE)

(Expanded Conceptual + Numerical Explanation)

◆ SECTION 1: CONCENTRATION, SOLUBILITY, HENRY (Q1–Q30)

Q1

Urea moles = $10/60 = 0.167$

Sucrose moles = $34.2/342 = 0.10$

Water moles = $100/18 \approx 5.56$

Total solute moles = 0.267

Total moles = $5.56 + 0.267 = 5.827$

$$X_{\text{solute}} = \frac{0.267}{5.827} \approx 0.046$$

✓ Answer: B (≈ 0.04)

Q2

$\text{Mass \%} = \frac{5}{5+95} \times 100 = 5\%$

✓ Answer: A

Q3

$$m = \frac{0.5}{0.5} = 1.0 \text{ } m$$

✓ Answer: B

Q4

$$M = \frac{0.5}{0.25} = 2.0 \text{ } M$$

✓ Answer: D

Q5

20% w/w $\Rightarrow$ 20 g NaOH in 100 g solution

Volume = $100 / 1.2 \approx 83.3$ mL

Moles NaOH = $20/40 = 0.5$

$$M = \frac{0.5}{0.0833} \approx 6.0$$

✓ Answer: B

Q6 (Henry's law)

$$C \propto P \Rightarrow C_2 = 3 \times 0.1 = 0.3$$

✓ Answer: A

Q7

$$S_2 = 5 \times 0.02 = 0.1$$

✓ Answer: B

Q8

$$C = \frac{P}{k_H} = \frac{0.5}{100} = 0.005$$

✓ Answer: C

Q9

Higher $k_H \Rightarrow$ lower solubility

✓ Answer: B

Q10

$$S_2 = \frac{5}{2} \times 0.02 = 0.05$$

✓ Answer: B

Q11

Solubility $\uparrow \Rightarrow k_H \downarrow$

✓ Answer: A

Q12

$$x_A^{vap} = \frac{x_A p_A^\circ}{x_A p_A^\circ + x_B p_B^\circ}$$

Given vapour fraction = 0.5 $\rightarrow$
Solving gives:

$$p_A^\circ : p_B^\circ = 3 : 2$$

✓ Answer: C

Q13

10 g glucose in 100 g solution

Volume = $100 / 1.04 \approx 96$ mL

Moles = $10/180 = 0.056$

$$M \approx \frac{0.056}{0.096} \approx 0.55$$

✓ Answer: A

Q14

1 m $\Rightarrow$ 1 mol solute + 55.5 mol water

$$X \approx \frac{1}{56.5} \approx 0.017$$

✓ Answer: A

Q15

Mass % depends on **molar mass**

✓ Answer: D

Q16

2 molal $\Rightarrow$ 2 mol solute per 1 kg water

Water moles = $1000/18 \approx 55.5$

✓ Answer: B

Q17

0.001 M = 1 mmol L⁻¹

✓ Answer: B

Q18

$$34/58.5 \approx 0.58$$

✓ Answer: A

Q19

Exothermic $\rightarrow$ Le Chatelier $\rightarrow$ solubility $\downarrow$ with T

✓ Answer: B

Q20

Endothermic $\rightarrow$ solubility $\uparrow$ with T

✓ Answer: A

Q21

At saturation: dissolution rate = crystallization rate

✓ Answer: C

Q22

Supersaturated $\rightarrow$ metastable

✓ Answer: B

Q23

Common ion ↓ solubility

✓ Answer: B

Q24

$$S \propto P \Rightarrow S = 0.01$$

✓ Answer: C

Q25

Henry: solubility $\propto$ pressure

✓ Answer: C

Q26

Large $k_H \Rightarrow$ very low solubility

✓ Answer: B

Q27

Helium less soluble $\rightarrow$ avoids bends

✓ Answer: B

Q28

$$S = 0.21S$$

✓ Answer: A

Q29

Higher T $\rightarrow$ higher $k_H \rightarrow$ lower solubility

✓ Answer: B

Q30

Deviation $\rightarrow$ gas reacts or highly soluble

✓ Answer: B

◆ SECTION 2: RAOULT, IDEAL, AZEOTROPES (Q31–Q60)

I'll now give compressed but concept-clear explanations (JEE Adv style).

31. $P = 0.25(120) + 0.75(80) = 90 \rightarrow \mathbf{B}$
 32. $y_A = \frac{30}{90} = 0.33 \rightarrow \mathbf{C}$
 33. Linear $\rightarrow \mathbf{C}$ (80)
 34. $x_A = 1/4 \rightarrow \mathbf{A}$
 35. Ideal $\Rightarrow \Delta H=0, \Delta V=0 \rightarrow \mathbf{B}$
 36. Weak A–B $\Rightarrow$ positive $\rightarrow \mathbf{A}$
 37. Strong A–B $\Rightarrow$ negative $\rightarrow \mathbf{B}$
 38. Above line $\rightarrow \mathbf{A}$
 39. Below line $\rightarrow \mathbf{B}$
 40. Benzene–toluene $\rightarrow \mathbf{C}$
 41. Ethanol–water $\rightarrow \mathbf{A}$
 42. Acetone–chloroform $\rightarrow \mathbf{B}$
 43. Vapour = liquid composition $\rightarrow \mathbf{B}$
 44. Positive deviation $\rightarrow \mathbf{A}$
 45. Negative deviation $\rightarrow \mathbf{B}$
 46. Ethanol–water $\rightarrow \mathbf{B}$
 47. HCl–water $\rightarrow \mathbf{A}$
 48. Vapour = liquid $\rightarrow \mathbf{A}$
 49. Same composition prevents separation $\rightarrow \mathbf{B}$
 50. When solute behaves like solvent $\rightarrow \mathbf{B}$
 51. Henry's law $\rightarrow \mathbf{A}$
 52. $p^\circ = 30/0.25 = 120 \rightarrow \mathbf{C}$
 53. Vapour richer in more volatile $\rightarrow \mathbf{B}$
 54. Ends at azeotrope $\rightarrow \mathbf{C}$
 55. Strong H-bonding $\rightarrow \mathbf{A}$
 56. Similar size/polarity $\rightarrow \mathbf{B}$
 57. Ideal $\Rightarrow \gamma = 1 \rightarrow \mathbf{B}$
 58. $a = \gamma x \rightarrow \mathbf{A}$
 59. Fails badly $\rightarrow \mathbf{B}$
 60. Two equations $\rightarrow \mathbf{B}$
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◆ SECTION 3: COLLIGATIVE & VAN'T HOFF (Q61–Q100)

Key formulas used repeatedly:

$$\Delta T = iKm, \quad \pi = iCRT$$

61. $i = \frac{0.512}{0.512} = 1 \rightarrow \mathbf{B}$
 62. $i = 2 \Rightarrow \alpha \approx 0.5 \rightarrow \mathbf{B}$
 63. Complete dimer $\rightarrow 0.5 \rightarrow \mathbf{C}$
 64. $i = 0.75 \Rightarrow \alpha \approx 0.5 \rightarrow \mathbf{B}$
 65. Lower than expected $\rightarrow$ association $\rightarrow \mathbf{A}$
 66. Effective double $\rightarrow i=2 \rightarrow \mathbf{C}$
 67. $\pi = CRT \Rightarrow M \approx 2460 \rightarrow \mathbf{C}$
 68. NaCl gives double particles $\rightarrow \mathbf{C}/2 \rightarrow \mathbf{B}$
 69. Sensitive at dilute $\rightarrow \mathbf{A}$
 70. $M \approx 107 \rightarrow \mathbf{B}$
 71. $\text{Na}_2\text{SO}_4 \rightarrow 3 \text{ ions} \rightarrow \mathbf{C}$
 72. $i = 2.5 \rightarrow \mathbf{C}$
 73. $i = 1 + 2\alpha \rightarrow \mathbf{A}$
 74. Observed/expected $\rightarrow 1.8 \rightarrow \mathbf{A}$
 75. Linear through origin $\rightarrow \mathbf{A}$
 76. Double concentration $\rightarrow$ double $\pi \rightarrow \mathbf{C}$
 77. Any colligative $\rightarrow \mathbf{D}$
 78. Osmotic pressure $\rightarrow \mathbf{C}$
 79. Same $\pi \rightarrow \mathbf{B}$
 80. Hypertonic $\rightarrow$ cell shrinks $\rightarrow \mathbf{B}$
 81. Hypotonic $\rightarrow$ swell $\rightarrow \mathbf{A}$
 82. $i = 3 \Rightarrow 0.558 \rightarrow \mathbf{C}$
 83. Only if constants equal $\rightarrow \mathbf{B}$
 84. $18/180 = 0.1 \rightarrow \mathbf{A}$
 85. Relative lowering $\approx$ mole fraction $\rightarrow \mathbf{A}$
 86. $\propto$ mole fraction $\rightarrow \mathbf{C}$
 87. Volatile solutes $\rightarrow \mathbf{A}$
 88. Infinite dilution $\rightarrow \mathbf{A}$
 89. Both methods $\rightarrow \mathbf{C}$
 90. Smaller $\Delta T_f \rightarrow$ association $\rightarrow \mathbf{A}$
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◆ SECTION 4: MIXED HIGH-LEVEL (Q101–120)

101. Approaches azeotrope $\rightarrow \mathbf{C}$
102. Residue $\rightarrow$ less volatile $\rightarrow \mathbf{B}$
103. Higher than expected $\rightarrow$ dissociation $\rightarrow \mathbf{B}$

- 104. Lower $k_H \rightarrow$ higher solubility $\rightarrow$ **A**
- 105. Always obeys Raoult $\rightarrow$ **A**
- 106. CS₂-acetone $\rightarrow$ **D**
- 107. Straight line $\rightarrow$ **A**
- 108. More particles $\rightarrow$ **B**
- 109. Must account reaction $\rightarrow$ **B**
- 110. Bond breaking $\rightarrow$ **A**
- 111. Hydrostatic pressure $\rightarrow$ **A**
- 112. $\propto$ particles per volume $\rightarrow$ **B**
- 113. High pressure membrane $\rightarrow$ **C**
- 114. Same molality $\rightarrow$ **B**
- 115. Usually increases $\rightarrow$ **A**
- 116. Inter-ionic effects $\rightarrow$ **B**
- 117. $\gamma \rightarrow 1 \rightarrow$ **B**
- 118. $\gamma > 1 \rightarrow$ positive $\rightarrow$ **A**
- 119. $\gamma < 1 \rightarrow$ negative $\rightarrow$ **A**
- 120. Total questions = 120 $\rightarrow$ **C**